

ERIC GITANGU

Software Engineering Lead / Architect — Distributed Systems and AI/ML

Nairobi, Kenya · deveric@unicorns.run · +254 708 078 997

linkedin.com/in/ericgitangu · github.com/ericgitangu · resume.ericgitangu.com

PROFESSIONAL SUMMARY

Software engineering lead and architect with 10+ years shipping event-driven distributed systems across healthcare ML, fintech, energy access, and government-scale digital infrastructure. Architectural focus: CQRS and event sourcing, hexagonal / ports-and-adapters, domain-driven bounded contexts, backend-for-frontend, and observability-first operations — applied pragmatically to resource-constrained, multi-country African market problems. Founded Vishnu Systems' machine learning security platform for AI-assisted medical devices (SiMD / SaMD), integrating CNNs, RAG, and GNNs with LLM APIs (OpenAI, Anthropic) and regulatory compliance tooling. Led engineering across 7 African markets at Ignite / ENGIE, designing async payment infrastructure and instrumenting distributed-tracing and APM stacks. Co-built government-scale digital addressing for 2.3M Nairobi residents at Baw Bab. Hands-on with FHIR-adherent patient data systems via Unicorns (unicorns.run), a personal platform exploring digital health architecture patterns. Polyglot across Python, Java, Rust, TypeScript, Go, Kotlin, Swift, and Clojure — language choice follows the architectural decision.

TECHNICAL SKILLS

Architectural Patterns: CQRS, Event Sourcing, Saga Orchestration, Hexagonal / Ports & Adapters, Domain-Driven Design (bounded contexts, aggregates), Backend-for-Frontend (BFF), Functional Core / Imperative Shell, Railway-Oriented Programming, Actor Model (OTP), CRDTs, Transactional Outbox, Circuit Breakers

Distributed Systems & Event Infrastructure: Kafka, RabbitMQ, AWS Kinesis / SQS / SNS / EventBridge, Redis Streams; idempotent consumers, DLQs, backpressure, exactly-once semantics; microservices, service mesh, gRPC, REST / GraphQL APIs

APM, Observability & Distributed Tracing: OpenTelemetry (OTel), Prometheus, Grafana, Jaeger, ELK Stack (Elasticsearch / Logstash / Kibana), AWS X-Ray, CloudWatch; SLO / SLI definition, error budgets, on-call runbooks, incident response

AI / ML & LLM Integration: OpenAI, Anthropic Claude, Google Gemini APIs; RAG systems, CNNs, GNNs, Transformers; TensorFlow, PyTorch, Apache Airflow, MLflow; AI-powered chatbots, intelligent automation, end-to-end model deployment, production MLOps

Data & ETL: PostgreSQL, Redshift, MongoDB, DynamoDB, Redis; ETL / ELT data pipelines, data ingestion, data quality and data validation; feature engineering; Vertex AI + pgvector for retrieval

Cloud & Serverless: AWS (Lambda, Cognito, CDK), GCP (Cloud Run, Vertex AI), Azure (App Service, Functions); Docker, Kubernetes, Terraform IaC; GitHub Actions CI/CD, Git

Frontend & Mobile (Frontline-Worker UX): React / Next.js, Progressive Web Apps (PWA) for low-bandwidth / low-literacy contexts; Kotlin + Jetpack Compose (Android native); Swift / SwiftUI (iOS native); Electron (desktop)

Secure APIs & Compliance: OWASP Top 10, NIST SSDF, HIPAA, FHIR / HL7 (hands-on via Unicorns patient EMR), AES-256-GCM field encryption, OAuth2 / JWT, secure coding, data privacy

Polyglot Delivery: Python (FastAPI, Django), TypeScript / Node.js, Java (Spring Boot), Groovy / Grails, Rust (Axum, PyO3), Go, Kotlin, Swift, Scala, Clojure, Solidity, C / C++, C# / .NET, SQL

Domains: Healthcare ML (SiMD / SaMD), digital health, fintech (mobile money: M-Pesa, Airtel Money), energy access (PAYG solar), agritech, multi-tenant SaaS, multi-country African markets

EXPERIENCE

Team Lead — QA & Engineering (SDET & Platform)

May 2025 – Mar 2026

Ignite Energy Access (formerly ENGIE Energy Access) · Nairobi, Kenya · 7 African markets

- Designed event-driven async payment architecture using RabbitMQ-backed bounded contexts (Payment, Token, Subscription) to decouple synchronous bottlenecks in the PAYG lifecycle; eliminated peak-load timeouts across 7 African markets handling millions of daily transactions.
- Instrumented full distributed tracing and APM stack across Ruby / Node.js services using OpenTelemetry, Prometheus, Grafana, Jaeger, and ELK (Elasticsearch / Logstash / Kibana); defined SLOs and error budgets; cut mean-time-to-detection for payment anomalies from hours to under 5 minutes.

- Led shift-left quality transformation combining DDD bounded-context modeling with BDD / TDD discipline; built Playwright and Cypress E2E suites covering 900+ critical-path scenarios (payments, token generation, IoT provisioning, subscription lifecycle) across 7 markets; reduced regression cycles from 3 days manual to 45 minutes automated; unblocked release velocity from 2 to 12+ deploys / week. Line-managed 4 engineers across distributed teams (Kenya, Uganda, Tanzania) with 2 promoted to senior within tenure.
- Led Mobisol Rails 5→7.2 platform upgrade underpinning payment processing, PAYG token dispatch, IoT device diagnostics, subscription lifecycle, and OTA firmware; coordinated rollback from 5% staging to 100% production during incident response with zero customer-facing downtime.
- Built Redshift + Kafka data pipeline ingesting IoT sensor telemetry from PAYG solar devices for payment event analytics, giving finance and operations teams near-real-time reporting across markets and reducing manual reporting burden.

Director of Engineering / Sole Technical Founder

Jan 2024 – Aug 2025

Vishnu Systems, Inc. · Atlanta, GA (Remote) · AI-Assisted Healthcare Devices and Software

- Founded and shipped a machine learning security and robustness platform from zero for Software in Medical Devices (SiMD / SaMD), where model failure has patient-safety implications; applied event-driven pipelines with idempotent workers and DLQs to guarantee evaluation integrity.
- Architected concurrent request-routing and fan-out backend: Python (FastAPI) orchestration layer, Rust + PyO3 evaluation core for performance-critical paths, Go middleware for rate limiting and channel-based worker coordination, GPU-backed evaluation workers on AWS.
- Shipped ML pipelines combining CNNs, RAG, and GNNs to detect adversarial attacks on medical ML models; orchestrated with Apache Airflow; tracked with MLflow for experiment versioning, model registry, and reproducibility; integrated LLM APIs (OpenAI, Anthropic) for compliance documentation and automated regulatory reporting.
- Built compliance tooling bridging data-science outputs to emerging FDA regulatory requirements for ML in medical devices — addressing a category gap with few established vendors.
- Operated full MLOps stack (TensorFlow, PyTorch, SQS / SNS / Kinesis for async messaging, hosted Kafka for high-throughput pipelines); on-call for production evaluations with customer-facing SLAs.

Lead Software Engineer

Jan 2023 – Aug 2025

Baw Bab Technologies, Inc. · Seattle, WA / Nairobi · Government-scale digital addressing

- Led 5-engineer team building digital addressing infrastructure for last-mile delivery in African urban areas where formal addresses don't exist — directly serving underserved communities and frontline field agents.
- Shipped Nairobi County Government pilot serving 2.3M residents — second digital addressing program of its kind in Africa after Rwanda's national deployment; integrated with county land registry for parcel-level interoperability with government systems.
- Designed computer-vision + GIS pipeline using high-resolution satellite imagery (Azure, Google, OpenStreetMap) and open-source vision models to identify residential structures and convert to Google Plus Codes at scale; optimized for low-bandwidth field operations.
- Owned full-stack delivery: Node.js application backend (Express) handling REST APIs, authentication, business logic, and React BFF orchestration; React frontend with mobile-first UX for field-agent workflows; decoupled specialist services for ML inference (Google Cloud Vision) and GIS pipeline, integrated via Google Cloud Pub/Sub event bus; MongoDB for spatial records; Kafka for geospatial event streaming; multi-cloud deployment (GCP Cloud Run + Azure) production-hardened for government-scale data workflows.

Senior Software Engineer

Jan 2022 – Dec 2023

Reinsurance Group of America (RGA) · Remote · Global insurance technology

- Delivered Java / Spring Boot microservices for actuarial, billing, and reinsurance treaty processing underpinning RGA's global operations; applied defensive coding and rigorous integration testing against PostgreSQL-backed services in a correctness-critical domain.
- Contributed to polyglot backend modernization across Groovy / Grails (legacy reinsurance services) and Node.js (newer integration and orchestration tiers), as part of a small team modernizing the platform service by service while preserving backward compatibility for downstream actuarial and treaty-processing consumers.
- Built React frontend for actuarial portals and treaty-processing UIs paired with the modernized backend services; maintained responsive UX across component-based frameworks.

- Profiled and resolved performance bottlenecks across reinsurance treaty processing, billing reconciliation, and actuarial reporting workflows; identified deprecated services for retirement and planned migration paths for dependent systems without functional regression.
- Refined service boundaries and data contracts with architects; operated in distributed team across US / EMEA time zones; owned CI / CD pipelines and release coordination for services in scope — small-team context meant end-to-end SDLC ownership rather than handoff to separate ops.

Founder & Principal Consultant

Jan 2019 – Feb 2022

Deveric Technologies · Nairobi, Kenya

- Founded technical consultancy and mentorship initiative serving early-career African developers; ran structured 1:1 mentorship (code reviews, system design walkthroughs, career coaching) for 15+ junior engineers, with 6+ advancing to mid / senior roles at regional startups.
- Shipped production projects across Go, Rust, and Python / FastAPI in hackathon and open-source contexts; completed 80+ validated technical certifications across cloud (AWS, Azure, GCP), AI / ML (TensorFlow, PyTorch), blockchain (Solidity, Polygon, Stellar), and application security.

Software Engineer

Jan 2018 – Jan 2019

Vecima Networks Inc. · Peachtree City, GA · Petabyte-scale video streaming

- Wrote low-level C and embedded assembly for petabyte-scale streaming pipelines with deterministic store-and-replay (zero frame loss) powering video delivery for global cable operators, broadcasters, and OTT providers.
- Implemented reactive stream handling with event-driven non-blocking I/O at the systems level; designed fault-tolerant architecture with redundant data paths, graceful degradation, and self-healing recovery.

Software Engineer

Dec 2015 – Feb 2018

Veracode · Burlington, MA · Application Security Platform

- Contributed to J2EE → microservices decomposition using Spring Boot, JPA, and Hibernate; designed HATEOAS-compliant REST APIs with Swagger documentation across refactored service boundaries; containerized services with Docker and orchestrated on Kubernetes clusters on AWS.
- Built CI / CD and DevSecOps integrations (Jenkins, GitLab) connecting Veracode's SAST / DAST / SCA scanning engine to enterprise toolchains (Jira, Visual Studio plugins) used by Fortune 500 customers.
- Shipped security-reviewed code aligned to OWASP Top 10 and NIST SSDF using C# / .NET, Python, and Core Java — secure-coding discipline applied to every subsequent role, including sensitive data handling in healthcare and fintech.

Software Engineer (Co-op)

Sep 2013 – May 2014

IBM · Littleton, MA · IBM Web Experience Factory

- Built automated testing architecture for IBM Web Experience Factory (formerly WebSphere Portlet Factory) using Java on Linux — applied test-automation-first discipline that shaped every subsequent engineering role.

SELECTED PROJECTS

Vunachain (vunachain.com) — AgriTech Contract Farming & Traceability — Personal technical advisory project: event-driven serverless backend for contract farming compliance and transparent farmer payouts. Rust Lambda microservices on AWS ARM64 (Graviton), Amazon EventBridge as event bus for farming lifecycle events (planting, inspection, harvest, delivery, payment), SQS with DLQs, S3, RDS PostgreSQL, CloudWatch and AWS X-Ray observability. Blockchain-anchored traceability via Solidity smart contracts on Celo with Merkle-tree batch anchoring. React / TypeScript dashboard and mobile-first PWA for field data collection.

Unicorns (unicorns.run) — Multi-tenant super-app for African SMBs with HIPAA-compliant, FHIR-adherent healthcare vertical (patient EMR, vitals, medications, AI-generated clinical summaries via Claude). Rust Lambda microservices on ARM64, Next.js PWA, Kotlin + Jetpack Compose Android, SwiftUI iOS, Electron desktop. RAG chatbot with human escalation. M-Pesa / Airtel Money / Stripe payments with Merkle-tree-anchored audit trails. Demonstrates end-to-end AI-enabled digital health application development for resource-constrained markets.

Vishnu Systems ML Security Platform — Production event-driven ML evaluation platform for medical-device compliance. Python + Rust / PyO3 + Go polyglot backend, CNN / RAG / GNN evaluation workers on AWS GPU instances, Airflow orchestration, MLflow registry, Kafka pipelines. Full architectural ownership.

PawaCloud (pawacloud-web.fly.dev) — Multilingual AI cloud advisory platform. FastAPI + Gemini 2.5 Flash backend with Rust + PyO3 text processing, HMAC-signed bearer tokens, PostgreSQL (Neon) session persistence, SSE streaming; Next.js 16 PWA frontend; deployed to GCP Cloud Run (africa-south1) with OpenTelemetry tracing; RAG via Vertex AI + pgvector.

LendStream (nanolend-app.vercel.app) — Polyglot micro-lending platform applying CQRS and event sourcing. Java 21 / Spring Boot command side, Clojure functional scoring engine, Kafka event backbone, transactional-outbox for reliable cross-context publishing; dual Kafka / QStash transport abstraction; deployed scale-to-zero on GCP Cloud Run + Vercel.

EDUCATION

B.S. Computer Science • Minor: Mathematics

2011 – 2015

University of Massachusetts Lowell

- Dean's List (multiple semesters) — Fall 2010, Spring 2011, Fall 2011, Spring 2012, Spring 2013, Fall 2013, Spring 2014, Fall 2014, Spring 2015.
- Honor Society: Omicron Delta Kappa (ODK) — national leadership honor society recognizing scholarship and leadership across academic and co-curricular achievement.
- Scholarships: Microsoft Scholarship Recipient (2012–2013) — awarded for exemplary academic performance in Computer Science; Google Scholarship Recipient (2011–2012) — awarded for academic performance and demonstrated potential in Computer Science.

SELECTED ACHIEVEMENTS

- **Open source:** 20+ production-ready public repositories on [GitHub](https://github.com) across Rust, Go, TypeScript, Python, and Java.
- **Mentorship & Teaching:** Founded Deveric mentorship initiative (2019–present) with 15+ engineers mentored and 6+ advanced to mid / senior roles; Volunteer Student Teacher at CodePath.org (May–Aug 2024), coaching the Advanced TIP103 cohort across fundamentals, data structures, and advanced algorithms (Certificate of Recognition awarded). Technical writing at developer.ericgitangu.com.
- **Certifications:** 80+ validated technical certifications across AWS, Azure, GCP, AI / ML, blockchain (Solidity / Polygon / Stellar), and application security (full list on [LinkedIn](https://www.linkedin.com/in/ericgitangu)).